

SOUNDARYA MAHADEV

SENIOR SOFTWARE ENGINEER

Embedded & Automotive Software | C/C++ |
AUTOSAR | AI-Driven Test Automation

CI/CD | Functional Safety | Cybersecurity

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PROFESSIONAL SUMMARY

Senior Software Engineer focused on embedded and automotive software, C/C++, AUTOSAR, software testing, AI-assisted test generation, CI/CD, code coverage, software validation, and engineering automation. Experienced in building reusable engineering solutions that analyze source code, generate unit tests, compile and execute tests, measure coverage, and support safety-oriented validation. Experience includes AUTOSAR Basic Software concepts and functional-safety practices aligned with ISO 26262, with additional work in automotive cybersecurity and OT security.

CORE TECHNICAL SKILLS

Languages: C, C++, Python, Java (OOPS), SQL,

Embedded / Automotive: Embedded Systems, Automotive Software, AUTOSAR, Basic Software

Testing: Unit Testing, Test Automation, AI-Assisted Test Generation, Unity Test Framework, Code Coverage, Software Validation

CI/CD & DevOps: Jenkins, GitHub Actions, Docker, Git, CMake, CI/CD Design

AI / Automation: LLM Integration, Gemini API, Source-Code Analysis, Engineering Automation, Gherkin/BDD

Safety & Security: Functional Safety, ISO 26262, ASIL-oriented Validation, MC/DC-oriented Validation, Automotive Cybersecurity, OT Security

PROFESSIONAL EXPERIENCE

REQUISIMUS — Senior Software Engineer | Bengaluru | March 2026 - Present

- Develop embedded and automotive software using C/C++ and related engineering tools.
- Develop automated testing approaches covering unit testing, test execution, code coverage, and continuous testing.
- Develop AI-assisted solutions for source-code analysis and automated unit-test generation.
- Build reusable frameworks that analyze source code, generate tests, build software, execute test suites, and collect coverage and validation results.
- Integrate automated build and testing workflows into CI/CD pipelines using Git-based development practices.
- Work with AUTOSAR Basic Software concepts and embedded software validation; apply software-quality and functional-safety concepts aligned with ISO 26262 and ASIL-oriented validation.
- Develop technical proof-of-concepts and engineering automation solutions across embedded software, AI-assisted testing, and CI/CD.

STRATACACHE — Content Promotion Specialist / Software Engineer | Bengaluru | December 2023 - March 2026

- Worked on software development, application maintenance, debugging, testing, validation, and delivery.
- Developed and maintained software components based on project requirements and collaborated with cross-functional teams.
- Worked with software development tools and version-control practices throughout the development lifecycle.
- Supported content-promotion activities using CMS tools and worked on promotional initiatives for multiple brands.

VIRTUSA — Associate Software Engineer | Bengaluru | February 2022 - December 2022

- Supported application development, testing, debugging, software maintenance, defect analysis, and issue resolution.
- Developed and maintained software components based on functional requirements and collaborated with team members throughout the software lifecycle.

- Followed structured development, version-control, and testing practices.

SELECTED REQUISIMUS PROJECTS

VERISAFE — AI-Assisted CI/CD Test Automation for Embedded C

- Designed and built an AI-assisted unit-test generation and CI/CD verification engine for embedded C codebases, evolved through five release milestones (v1.0.0 → v3.1.3).
- Built static-analysis classification for source files and LLM-drafted Unity unit tests; every draft is compiled with a real toolchain and executed before being trusted.
- Packaged the engine as a standalone PyInstaller executable, portable Docker image published via GitHub Container Registry, and parameterized multi-client Jenkins CI/CD pipeline with Gherkin/BDD front end and JSON configuration.
- Built a release watcher that polls the registry approximately every five minutes and automatically triggers the verification pipeline when a new build ships, deliberately avoiding inbound webhook exposure.
- Root-caused and fixed AI-drafted assertion errors, a stale-manifest reporting bug, and Jenkins/Docker-outside-of-Docker infrastructure issues.
- Validated on an NXP S32K144-based smart-HMI automotive ECU codebase: 40/40 generated tests passing, 99.0% line coverage, and 97.4% branch coverage, reproduced across executable, Docker, and Jenkins environments.

Technology: Python, Docker, Jenkins, GitHub Actions, C/Embedded Systems, Unity Test Framework, Gemini API/LLM Integration, Gherkin/BDD, Git

AUTOSAR Basic Software CI/CD Demonstrator

- Developed a demonstration environment for an AUTOSAR-oriented Basic Software stack with automated build, testing, coverage analysis, and CI/CD integration.
- Demonstrated AUTOSAR BSW modules including DIO, Port, DET, and ECU Manager, together with embedded C testing and automated unit testing.
- Integrated CI/CD workflow execution, code coverage, AI-assisted software analysis, and automated reporting to demonstrate a repeatable embedded software validation workflow.

Technology: AUTOSAR BSW, Embedded C, Unit Testing, CI/CD, Code Coverage, AI-Assisted Analysis, Automated Reporting

Multiagent Orchestration Platform

- Built a platform for creating AI agents and wiring multiple agents into one workflow through a visual diagram rather than code.
- Implemented a FastAPI + React application backed by PostgreSQL and Qdrant, with six Docker services.
- Enabled agents to answer from user documents and run on OpenAI, Anthropic, Google, or a local model without an API key or data leaving the machine.
- Made workflows accessible through a browser, REST call, webhook, or Slack mention; CI included 418 backend and 95 frontend tests.

Technology: FastAPI, React, PostgreSQL, Qdrant, Docker, OpenAI, Anthropic, Google models, REST, Webhooks, Slack

VERISAFE — AI-Assisted C/C++ Software Testing & CI/CD Framework

- Developed a proof-of-concept framework for automating software testing activities for C/C++ projects.
- Implemented source-code analysis, AI-assisted unit-test generation, automated build and test execution, code coverage analysis, CI/CD integration, safety-oriented validation, ASIL-based coverage validation, MC/DC-oriented validation for higher safety levels, and automated test/compliance reporting.
- Designed the workflow to reduce manual test-development effort and provide a reusable approach for continuous software quality and validation.

Technology: C/C++, AI-Assisted Testing, CI/CD, ASIL, MC/DC, Code Coverage, Software Validation

EDUCATION

Sri Tontadarya College of Engineering, Gadag — Bachelor's Degree, Electrical, Electronics and Communications Engineering | Aug 2016 - Jul 2021

Indu Pre University College — 12th, Science | Jun 2014 - Apr 2016

Gurudev English Medium High School — 10th | Jun 2013 - Apr 2014

CERTIFICATION

- Oracle Certified Professional, Java SE 8 Programmer II